

0914 U1 investigate questions

1. A city's annual car auction sells vintageolder vehicles. A car collector was curious to know whether the mean selling price of the vehicles was different between the auction years 2024 and 2026. Approximately 2,000 vehicles were sold at each auction, and none of the vehicles were sold at both auctions. The collector took a random sample of 36 vehicles sold at the 2024 auction and calculated a mean selling price of \$106,400 and a standard deviation of \$32,000. The collector also took a random sample of 42 vehicles sold at the 2026 auction and calculated a mean selling price of \$96,500 and a standard deviation of \$22,000.

Complete the appropriate inference procedure at the 98% confidence level to estimate the difference between the population means of the vehicle selling prices for 2024 and 2026.

Respond to parts A, B, C, D, and E.

Part B

Identify the type of study being conducted.

Part B Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- The type of study the researcher conducts is an observational study.



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The student response accurately **identifies** the type of study being conducted.

Responses that earn this point:

- The response accurately identifies an observational study.

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A country's transportation department wants to conduct a survey to collect data from adults, ages 18 and older, in that country. The survey will be focused on commuting to work, including commute time in minutes, method of transportation used, and other information.

Country A is divided into many regions, and each region will be asked to send out a survey and collect the data. The transportation department will allow each region to use any method it prefers to survey the adults.

Three different regions will use the following sampling methods:

Region X: Each adult who lives in the region will receive a survey and will be required to complete the survey by mail. Adults who do not complete the survey will be sent reminders to do so until they have completed the survey.

Region Y: The region is divided into six areas that are similar to each other. Two of the six areas will be randomly selected, and each adult who lives in those two areas will receive a survey and be asked to complete the survey by mail. The table gives the results of the data collection for the two areas in Region Y for the question "What is your average daily commute time to work in minutes?"

Region Z: Each adult who lives in the region and has responded to a previous official survey will receive a survey and be asked to complete it by mail.

Summary Statistics

Area	$\bar{x}$	n	Min	Q1	Median (Q2)	Q3	Max
One	55	2,887	7	35	53	74	101
Five	42	4,502	5	24	32	68	83

2. Respond to parts A, B, C, D, and E, and any subparts.

Part E

Manuel wants to perform a study about the average daily commute time to work for Area Three and Area Six in Region Y. He believes there is a difference in the commute times for the two areas. **Determine** a valid investigative question for Manuel's study. Ensure all three components of a valid investigative question are included in your response.

Part E Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- Is there convincing statistical evidence to suggest that the mean daily commute time to work for all adults (ages 18 and older) in Area Three of region Y is different than the mean daily commute time to work for all adults (ages 18 and older) in Area Six of region Y?

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The student response accurately **determines** the first component of an investigative question.

Responses that earn this point:

- The response accurately reports the variables as the two areas and daily commute times.
- The response accurately reports the variables as Areas Three and Six and daily commute times.

Examples that earn this point:

- The two areas and mean daily commute times
- Areas Three and Six and average daily commute times

Part E Point 2

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- Is there convincing statistical evidence to suggest that the mean daily commute time to work for all adults (ages 18 and older) in Area Three of region Y is different than the mean daily commute time to work for all adults (ages 18 and older) in Area Six of region Y?



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The student response accurately **determines** the second component of an investigative question.

Responses that earn this point:

- The response accurately reports the population parameters as the mean daily commute time for each area and the direction of alternative hypothesis as different (difference).
- The response accurately reports the population parameters as the mean daily commute time for each area and the direction of alternative hypothesis as not the same.

Examples that earn this point:

- Mean daily commute times for Areas Three and Six are different (difference)
- Mean daily commute times for Areas Three and Six are not the same.

Part E Point 3

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.

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- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- Is there convincing statistical evidence to suggest that the mean daily commute time to work for all adults (ages 18 and older) in Area Three of region Y is different than the mean daily commute time to work for all adults (ages 18 and older) in Area Six of region Y?



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The student response accurately **determines** the third component of an investigative question.

Responses that earn this point:

- The response accurately reports that the population is all adults in Areas Three and Six of Region Y.

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3. Respond to parts A, B, C, D, and E, and any subparts.

Part B

- i. **Explain** why the results of the sampling method for Region Y can be generalized to the population of the region.
- ii. Tom, a member of the planning committee for Region Z, proposed Region Z's sampling method. Tom claims the sampling method for Region Z may lead to a higher survey response rate than the sampling methods used for the other two regions and could better represent the entire population in Region Z. **Explain** why Tom's claim is not correct.

Part B Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- The findings would represent the entire region since the 6 areas are similar to each other and data from two randomly selected areas are collected.



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The student response accurately **explains** why a method is appropriate or not appropriate for gathering and representing data.

Responses that earn this point:

- The response accurately explains that the method for Region Y is appropriate because the two areas were randomly selected and could represent the entire region since the 6 areas are similar to each other.

Part B Point 2

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- Tom's sampling method is not better since it is not a random sample and also may not represent the population of the region because he is only surveying people who responded to a previous survey.



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The student response accurately **explains** why a method is appropriate or not appropriate for gathering and representing data.

Responses that earn this point:

- The response explains that Tom's sampling method is not better because it is not a random sample and may not represent the population of the region.
- The response explains that the survey was only sent to people who had previously responded to a survey and not randomly selected, so the results will not represent all the registered voters in that region.

Examples that earn this point:

- The results cannot be generalized to all registered voters in Region Z because the survey was not sent to people who were randomly selected and was only sent to people who previously responded to a survey, so the results will not represent all the registered voters in that region.
- The adults who live in this region were not randomly selected, so the results will not represent all the registered voters in that region.

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4. Andrea, a cybersecurity specialist at a bank, was concerned about the security of the bank's website. She was asked to compare two encryption processes where information is scrambled and cannot be easily read methods that could be used to protect the website's banking information. Her department was concerned that the older encryption version, DES, may be weaker than the newer encryption version, AES. One measure of how well an encryption method works is session duration, how long the information is accessed in seconds. A very long session duration may indicate that there has been unauthorized access to the data, which means that the encryption method needs to be more secure. Her department wanted her to study the two encryption methods.

Therefore, Andrea conducted a study of the two types of encryption methods. Andrea randomly collected this information from bank customers who used the website over the previous six months. For each session of the sampled customers, Andrea collected the session duration, in seconds, and the type of encryption method used. The table displays the summary statistics for her study.

Summary Statistics

Encryption Method	n	$\bar{x}$	s	Min	Q1	Median (Q2)	Q3	Max
DES	2,865	782	768	1	226	543	1,107	6,950
AES	4,706	794	803	1	233	794	1,091	7,190

Respond to parts A, B, C, D, and E, and any subparts.

Part E

Determine the population for a valid investigative question for Andrea's study.

Part E Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution

- All the bank's customers who use the website



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The student response accurately **determines** the indicated component for a valid investigative question.

Responses that earn this point:

- The response accurately determines that the population for a valid investigative question for Andrea’s study is all the bank’s customers who use the bank’s website.

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5. A city’s public health department recently started a “Hydration and Health” campaign to encourage adults to drink enough water each day. In the city, doctors have reported that many adults substitute caffeinated beverages, such as coffee and tea, for water. They are worried this might keep adults from staying properly hydrated.

To look into this issue, the department surveyed a random sample of 300 adults from across the city. Each person filled out a short survey about their beverage preference and whether they meet the daily water intake recommendation. The following variables are used to create a table and mosaic plot.

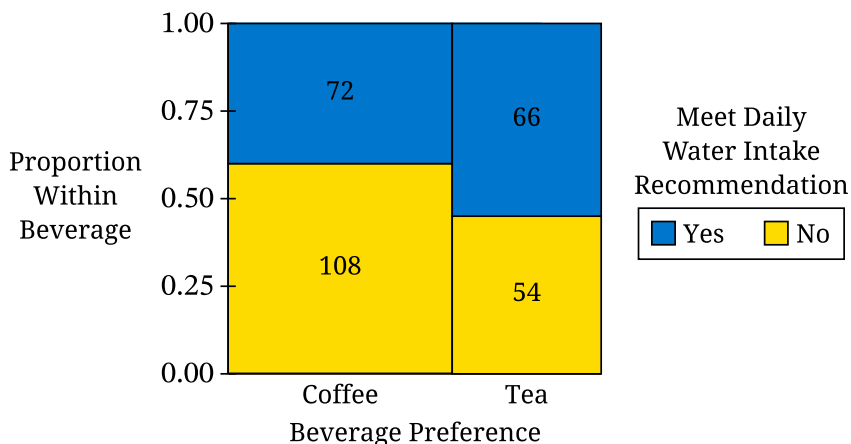
- Beverage Preference: Coffee or Tea
- Meets Water Intake Recommendation: Yes or No

The department plans to use the survey results to help create educational materials for the campaign. City officials also hope the information will show whether new ideas—like adding free water refill stations—might help adults who are not drinking enough water. The survey results from the 300 residents are shown in the table.

Beverage Preference and Meets Water Intake Recommendation

		Meet Daily Water Intake Recommendation		
		Yes	No	Total
Beverage Preference	Coffee	72	108	180
	Tea	66	54	120
	Total	138	162	300

Beverage Preference and Meets Water Intake Recommendation



Respond to parts A, B, C, D, and E, and any subparts.

Part E

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The public health department is placing free water refill stations throughout the city to study whether doing so increases the proportion of adults across the city who meet the daily water intake recommendation. **Determine** a valid investigative question for the public health department's study. Ensure all three components of a valid investigative question are included in your response.

Part E Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- For all adults in the city, does placing free water refill stations throughout the city increase the population proportion who meet their daily water intake compared to not placing free water refill stations throughout the city?



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The student response accurately **determines** the first component of an investigative question.

Responses that earn this point:

- The response accurately reports the variables as the placement of free water refill stations throughout the city and adults in the city who meet their daily intake of water.

Examples that earn this point:

- For all adults in the city, is the presence of free water refilling stations throughout the city associated with a higher population proportion who meet their daily water intake compared to not having free water refill stations throughout the city?

Part E Point 2

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- For all adults in the city, does placing free water refill stations throughout the city increase the population proportion who meet their daily water intake compared to not placing free water refill stations throughout the city?



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The student response accurately **determines** the second component of an investigative question.

Responses that earn this point:

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- The response accurately reports the population parameters as the population proportion who meet their daily water intake and the direction of alternative hypothesis as increase.

Examples that earn this point:

- For all adults in the city, is the presence of free water refilling stations throughout the city associated with a higher population proportion who meet their daily water intake compared to not having free water refill stations throughout the city?

Part E Point 3

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- For all adults in the city, does placing free water refill stations throughout the city increase the population proportion who meet their daily water intake compared to not placing free water refill stations throughout the city?



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The student response accurately **determines** the second component of an investigative question.

Responses that earn this point:

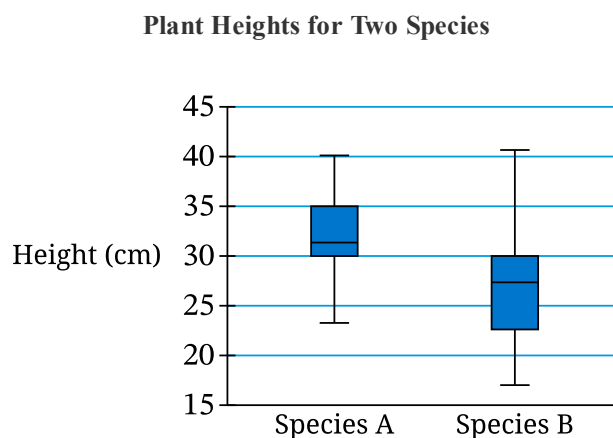
- The response accurately reports that the population is all adults in the city.

Examples that earn this point:

- For all adults in the city, is the presence of free water refilling stations throughout the city associated with a higher population proportion who meet their daily water intake compared to not having free water refill stations throughout the city?

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6. A team of biologists is investigating how two plant species commonly used in ecological restoration process of helping damaged or destroyed ecosystems regain their natural structure and function—Species A, a native grass, and Species B, a wildflower—respond to limited water availability. Although both species typically grow at the same rate under similar soil conditions, their growth may differ when water is limited. To study this possibility, the biologists randomly assigned Species A seeds to 35 plots and Species B seeds to 35 plots that were located in a controlled greenhouse environment with restricted watering. After 12 weeks, they recorded the height, in centimeters (cm), of every plant. The data is displayed for the resulting height distributions of both species.



Respond to parts A, B, C, D, and E, and any subparts.

Part E

The team of biologists plans to conduct a larger follow-up study to determine whether Species A plants grow taller when they receive a limited amount of water than when they receive a normal amount of water. **Determine** the alternative hypothesis the team should use for their investigative question.

Part E Point 1

To earn this point, responses must demonstrate the following:

- Except where otherwise noted, each point of these scoring guidelines is earned independently.
- The scoring criteria identify the specific components of the model solution that are used to determine the score.

Sample Solution:

- The team of biologists would have the following alternate hypothesis: The true mean height, in centimeters, for Species A plants exposed to limited water conditions will be greater than the true mean height, in centimeters, for Species A plants exposed to normal water conditions.



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The student response accurately **determines** the indicated component for a valid investigative question.

Responses that earn this point:

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- The response accurately determines the alternate hypothesis: The true mean height, in centimeters, for Species A plants exposed to limited water conditions will be greater than the true mean height, in centimeters, for Species A plants exposed to normal water conditions.
- The response accurately determines the alternate hypothesis: The true mean height, in centimeters, for Species A plants exposed to normal water conditions will be shorter than the true mean height, in centimeters, for Species A plants exposed to limited water conditions.